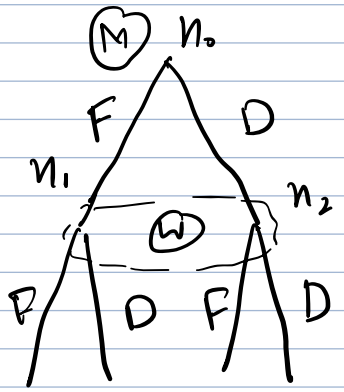
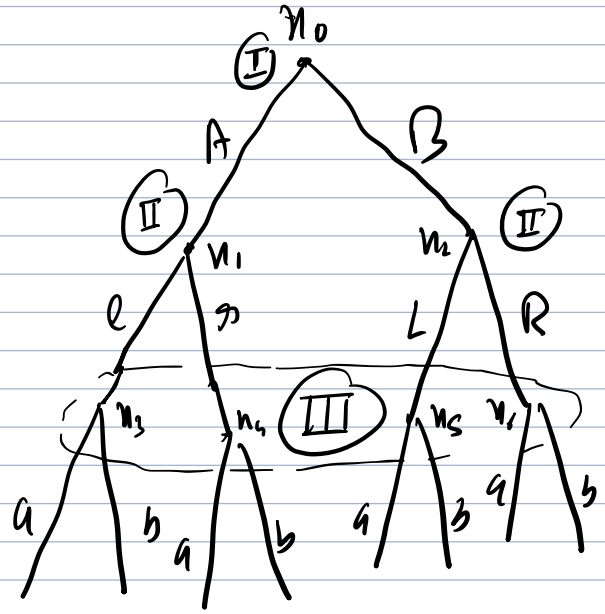
# Class 27



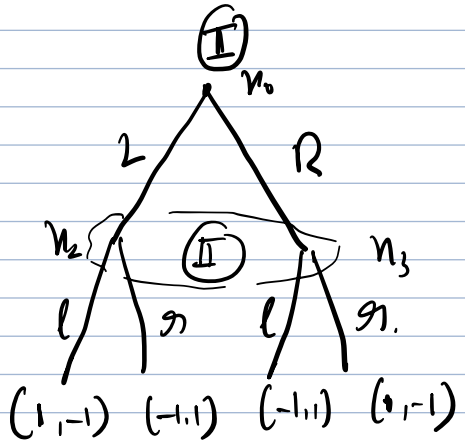

**Defn** For a game $\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N \cup \{0\}}, \mathcal{H}, (A(h))_{h \in \cup H_i}, (p_n)_{n \in V_0}, (u_i)_{i \in N} \rangle$

a subgame starting at a node $y \in V$ is a game $\Gamma_y = \langle N, V_y, E_y, y, (V_i)^y_{i \in N \cup \{0\}}, \mathcal{H}^y, (A(h))^y_{h \in \cup H_i}, (p_n)_{n \in V_0}, (u_i)_{i \in N} \rangle$

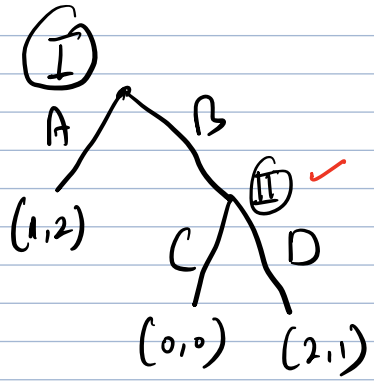
and

every information set $h$ must either be contained in $V_y$ or in $V \setminus V^y$.

**Defn** A strategy profile $s$ is an SPE if it is an NE in all the subgames



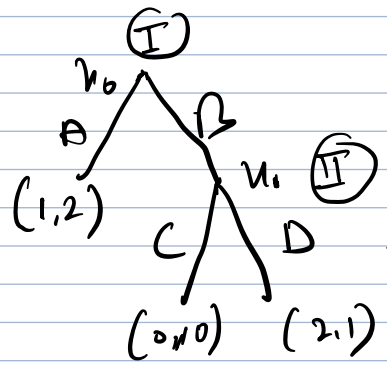

| | l | r |
|---|---|---|
| L | (1,-1) | (-1,1) |
| R | (-1,1) | (1,-1) |




Two NE, (A, C), (B, D)

One SPE (B, D).

- For a game $\Gamma$ and an NE $s$ of the game $\Gamma$, if a subgame is reached when players play the strategy profile $s$, then $s$ will constitute an NE for that subgame.




| | C | D |
|---|---|---|
| A | (1,2) | (1,2) |
| B | (0,0) | (2,1) |

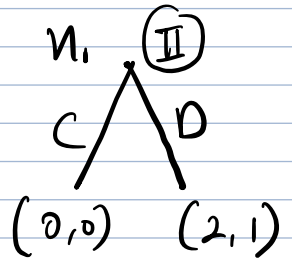
$(A, \alpha[C], (1-\alpha)[D])$

$\alpha \geq 1/2$

$(B, D)$

SPE $(B, D)$, $(A, \alpha[C], (1-\alpha)[D])$ ✓

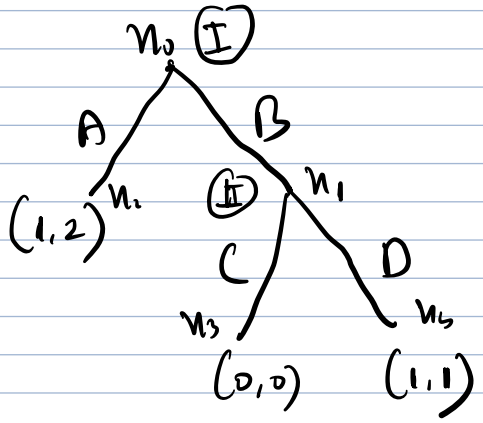
$1 > \alpha \geq 1/2$




$(\alpha[C], (1-\alpha)[D])$ is not an NE

only NE D

SPE $(B, D)$




| | C | D |
|---|---|---|
| A | (1,2) | (1,2) |
| B | (0,0) | (1,1) |

$(A, C)$, $(A, D)$ not SPE

$((\frac{1}{2}[A], \frac{1}{2}[B]), D)$ SPE

$(B, D)$ SPE

**Theorem**

Let $\sigma^*$ be an NE (either in mixed strategies or in behavior strategies) of a game $\Gamma$ and $\Gamma_n$ be a subgame of the game. If $P(n; \sigma^*) > 0$ then $\sigma^*$ is also an NE for the subgame $\Gamma_n$.